

Chapter 9 Critical Thinking

1. **What index value does the third element of an array have?**

Index Value 2

2. **Write the declaration for an array named quantities that stores 20 integers**

```
int[] money = new int[25];
```

3. **Write a declaration for an array named heights storing the numbers 1.65, 2.15, and 4.95.**

```
double[] heights = { 1.65, 2.15, 4.95 };
```

4. **How does passing an entire array to a method differ from passing a single element of the array?**

Passing an entire array allows the method to copy it while passing a single element only passes that value.

5. **Why are offset array indexes required in some cases?**

Because of the way offset arrays are stored.

6. **What output is displayed by the statements below?**

```
String name = "Elaine";  
System.out.println(name.charAt(3));
```

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7. **Give an example of when a dynamic array might be a better structure choice over an array**

Because dynamic arrays can grow during runtime or shrink. Which may be useful depending on what you're working on..

8. **How does the ArrayList indexOf() method determine equality between the object passed to the method and an element in the array.**

It does a search to check if the object has passed or if it has not passed.

9. **How can the values of wrapper class objects be compared?**

Wrapped class objects are used to encapsulate primitive data types.